

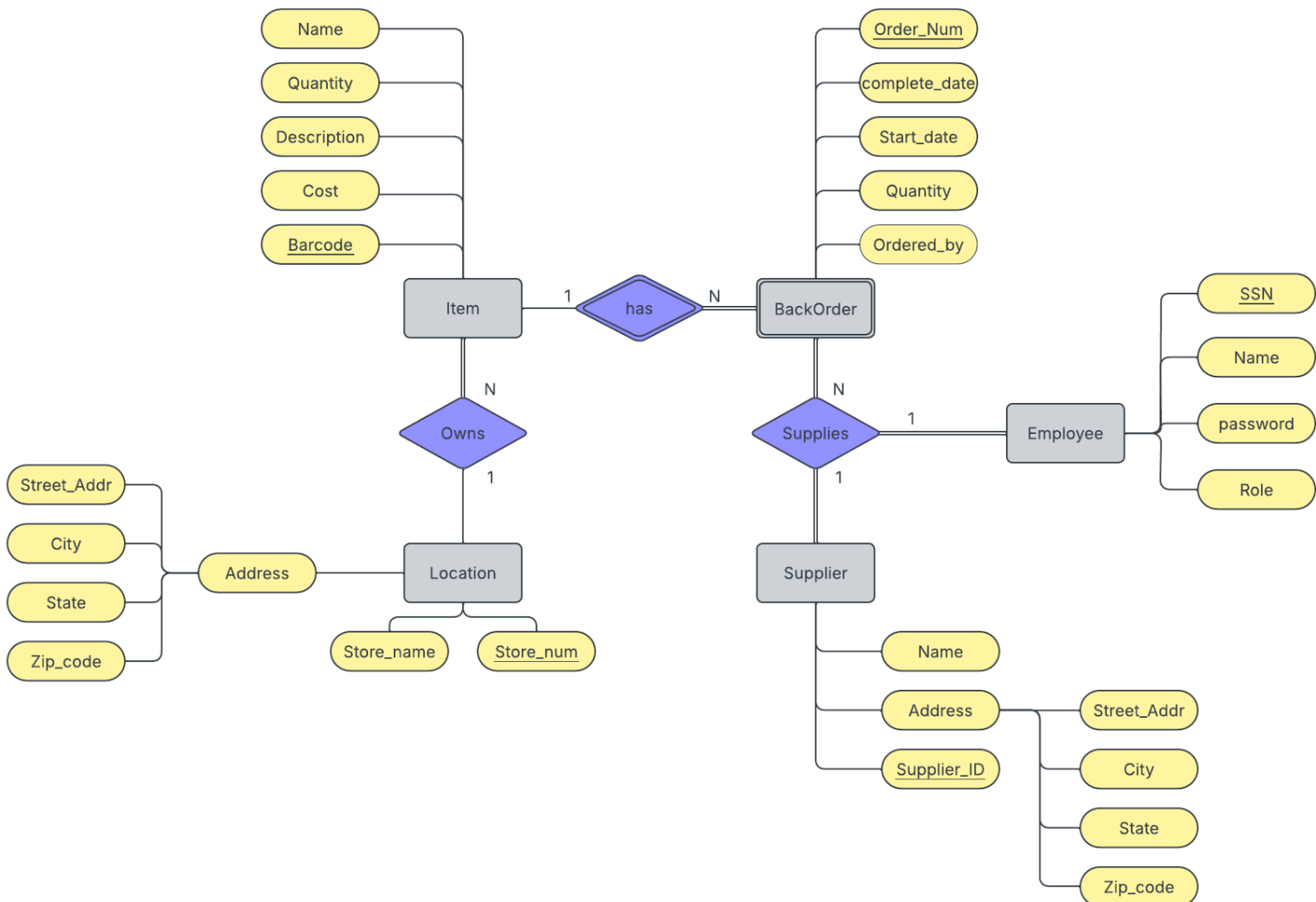
Project Phase II - Inventory Database

Connor Winning

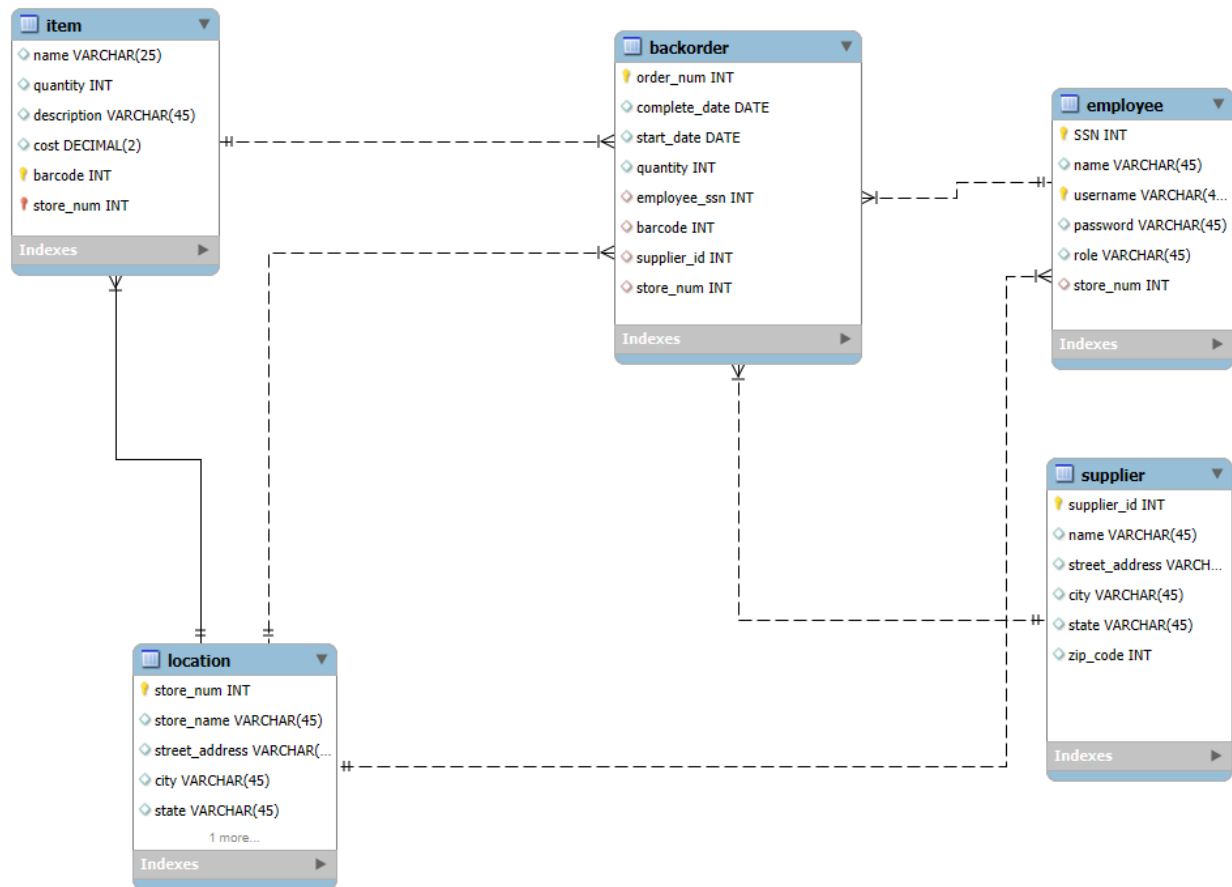
- **Problem Statement:**

My project is a perpetual inventory system that will keep track of the current inventory in a store. Every entity in the database should have a specific bar code that will be used as a key to search for. Items should be able to be added and removed from the inventory system. It must store the cost, name, quantity, and other information of every item in the store, and it must store back-order information about items currently ordered. Employee information should be stored for each location. Other information includes supplier information, supply order logging, and store location information. This will be done through a simple menu-style command-line interface.

- **Conceptual Database Design:**



- **Logical Database Design:**



- **Application Program Design:**

Login():

```
Username = prompt for username
Password = prompt for password
Query employee table
if( Check for username and password match ):
    Is_logged_in = true
if( employee is manager ):
    Is_admin = true
```

add_item():

```
Read JSON data from the front end
Name = JSON.get(name)
Quantity = JSON.get(Quantity)
Description = JSON.get(Description)
Cost = JSON.get(Cost)
Barcode = JSON.get(Barcode)
Store_num = Query employee table for store num by employee username
Add item data to the item table
```

create_backorder():

```
Order_num = generate_order_num()
Start_date = get_current_date()
Employee_username = get username from instance
Store_num = query employee table for store_num by username
Read JSON data from the front end
Complete_date = JSON.get(name)
Quantity = JSON.get(quantity)
Barcode = JSON.get(barcode)
Supplier_id = get selected supplier from front end
```

remove_item():

```
Do delete query in item table
```

remove_backorder():

```
Do delete query in backorder table
```

Remove_location():

if(not is_admin):

 Prompt user ("Admin Privlages Required for Action")

 Return

Do delete query in location table

Cascade to item table

Cascade to employee table

Cascade to backorder table

Add_location():

If (not is_admin):

 Prompt user ("Admin Privlages Required for Action")

 Return

Read JSON data from the front end

Store_name = JSON.get(store_name)

Address = JSON.get(address)

Store_num = generate_store_num

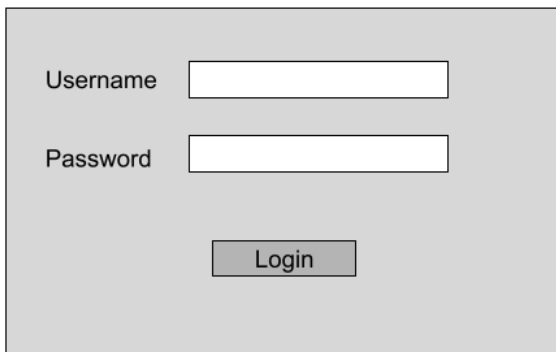
show_items():

store_num = query store_num from employee table

Items[] = query filtered data from the item table by store_num

Show items in the front end

- **Graphical User Interface Design:**



Username

Password

Login

<u>Item</u>		<u>Supplier</u>	
Item Name		Suppliers	
Quantity			(drop down menu) (select Supplier)
Cost			
Description			
Barcode			
			<input type="submit" value="submit"/>

<u>Item</u>		<u>Supplier</u>	
Barcode		Suppliers	
Order Date			(drop down menu) (select Supplier)
Est. Delivery Date			
Quantity			
			<input type="submit" value="submit"/>

